

## UST-KAMCHATSKY, Russian Federation

WMO: 324080

Lat: 56.2369N

Lon: 162.6919E

Elev: 30

StdP: 100.96

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-28.7 | (c)<br>-25.9 | (d)<br>-31.6                  | (e)<br>0.2 | (f)<br>-28.5 | (g)<br>-28.9 | (h)<br>0.3 | (i)<br>-25.5 | (j)<br>16.3           | (k)<br>-1.7 | (l)<br>14.5 | (m)<br>-1.8 | (n)<br>2.7            | (o)<br>350 | (p)<br>0.812 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>6.4             | (c)<br>20.7     | (d)<br>15.3 | (e)<br>18.7 | (f)<br>14.2 | (g)<br>16.9 | (h)<br>13.3 | (i)<br>16.0         | (j)<br>19.6 | (k)<br>14.8 | (l)<br>17.6 | (m)<br>13.8 | (n)<br>16.2 | (o)<br>3.8           | (p)<br>200 |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.4                            | 10.3 | 17.4 | 13.4 | 9.6 | 16.1 | 12.5 | 9.1 | 15.0 | 45.1          | 20.0 | 41.7 | 17.8 | 38.9 | 16.3 | 20.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 13.2              | 11.2 | 9.6 | -32.1                      | 24.7 | 2.8                | 2.0 | -34.1                                              | 26.1 | -35.7      | 27.2 | -37.3      | 28.3 | -39.3      | 29.7 |
|                   |      |     | -32.7                      | 18.2 | 2.9                | 1.3 | -34.8                                              | 19.1 | -36.5      | 19.8 | -38.1      | 20.5 | -40.2      | 21.5 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 0.2    | -11.8 | -11.2 | -7.1  | -2.8  | 2.0   | 8.0   | 12.1  | 12.7  | 9.3   | 3.2   | -3.3  | -9.5 |
|                                                                                  | DBStd       | 9.88   | 8.09  | 7.30  | 6.04  | 3.68  | 2.23  | 2.89  | 2.63  | 2.21  | 2.63  | 3.27  | 4.67  | 6.62 |
|                                                                                  | HDD10.0     | 3774   | 675   | 594   | 531   | 385   | 248   | 74    | 7     | 3     | 43    | 210   | 399   | 604  |
|                                                                                  | HDD18.3     | 6622   | 933   | 828   | 789   | 635   | 506   | 309   | 193   | 176   | 271   | 468   | 649   | 863  |
|                                                                                  | CDD10.0     | 194    | 0     | 0     | 0     | 0     | 0     | 14    | 72    | 86    | 22    | 0     | 0     | 0    |
|                                                                                  | CDD18.3     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 9      | 0     | 0     | 0     | 0     | 0     | 1     | 7     | 2     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 4.5   | 5.0   | 4.6   | 4.5   | 3.9   | 4.3   | 4.0   | 3.7   | 3.6   | 3.8   | 3.9   | 4.2  |
| Precipitation                                                                    | PrecAvg     | 701    | 79    | 62    | 55    | 36    | 29    | 35    | 40    | 64    | 59    | 74    | 74    | 86   |
|                                                                                  | PrecMax     | 868    | 151   | 130   | 91    | 67    | 66    | 80    | 76    | 120   | 182   | 133   | 132   | 161  |
|                                                                                  | PrecMin     | 459    | 12    | 6     | 13    | 11    | 5     | 1     | 1     | 20    | 13    | 34    | 24    | 42   |
|                                                                                  | PrecStd     | 85     | 39    | 33    | 23    | 16    | 14    | 20    | 21    | 29    | 38    | 30    | 30    | 36   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 2.6   | 2.1   | 2.6   | 4.9   | 12.2  | 21.7  | 24.2  | 22.7  | 18.4  | 11.7  | 7.2   | 4.2  |
|                                                                                  |             | MCWB   | 1.7   | 1.0   | 1.2   | 1.8   | 7.1   | 13.5  | 16.8  | 17.0  | 12.7  | 7.0   | 5.5   | 3.2  |
|                                                                                  | 2%          | DB     | 1.2   | 0.8   | 1.6   | 3.2   | 9.2   | 18.2  | 21.1  | 20.3  | 16.4  | 10.3  | 5.6   | 2.1  |
|                                                                                  |             | MCWB   | 0.5   | 0.1   | 0.5   | 0.6   | 5.3   | 11.8  | 15.8  | 15.6  | 12.0  | 7.0   | 4.4   | 0.9  |
|                                                                                  | 5%          | DB     | 0.4   | 0.0   | 0.9   | 2.1   | 7.1   | 15.6  | 18.9  | 18.4  | 15.0  | 9.2   | 4.4   | 0.9  |
|                                                                                  |             | MCWB   | -0.2  | -0.5  | 0.0   | 0.4   | 4.2   | 10.8  | 14.8  | 14.4  | 11.3  | 6.6   | 3.1   | 0.1  |
|                                                                                  | 10%         | DB     | -0.6  | -0.9  | 0.2   | 1.4   | 5.5   | 13.3  | 16.9  | 16.7  | 13.7  | 8.2   | 3.1   | -0.2 |
|                                                                                  |             | MCWB   | -1.1  | -1.5  | -0.6  | 0.0   | 3.2   | 9.7   | 13.7  | 13.8  | 10.9  | 6.1   | 1.7   | -0.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 1.7   | 1.2   | 1.7   | 2.4   | 7.5   | 14.0  | 17.6  | 17.9  | 14.1  | 9.2   | 6.0   | 3.2  |
|                                                                                  |             | MCDB   | 2.3   | 1.6   | 2.3   | 3.7   | 11.5  | 20.3  | 22.6  | 21.5  | 16.5  | 10.4  | 6.7   | 3.9  |
|                                                                                  | 2%          | WB     | 0.6   | 0.3   | 0.8   | 1.2   | 5.5   | 12.3  | 16.2  | 16.2  | 13.0  | 8.1   | 4.5   | 1.1  |
|                                                                                  |             | MCDB   | 1.0   | 0.6   | 1.3   | 2.3   | 8.6   | 17.1  | 20.3  | 19.3  | 15.0  | 9.3   | 5.4   | 1.7  |
|                                                                                  | 5%          | WB     | 0.0   | -0.4  | 0.2   | 0.7   | 4.5   | 11.1  | 15.1  | 15.0  | 12.1  | 7.2   | 3.2   | 0.2  |
|                                                                                  |             | MCDB   | 0.3   | 0.0   | 0.7   | 1.6   | 6.8   | 15.0  | 18.3  | 17.4  | 14.1  | 8.6   | 4.2   | 0.8  |
|                                                                                  | 10%         | WB     | -1.1  | -1.5  | -0.4  | 0.3   | 3.5   | 9.8   | 14.0  | 14.0  | 11.4  | 6.2   | 2.0   | -0.8 |
|                                                                                  |             | MCDB   | -0.6  | -1.0  | 0.1   | 1.2   | 5.3   | 13.0  | 16.6  | 16.1  | 13.2  | 7.9   | 2.9   | -0.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.3   | 7.6   | 7.6   | 6.2   | 4.8   | 6.6   | 6.6   | 6.4   | 7.5   | 7.0   | 6.6   | 7.3  |
|                                                                                  |             | MCDBR  | 4.9   | 5.0   | 4.7   | 6.3   | 8.2   | 12.0  | 11.0  | 10.2  | 9.2   | 8.2   | 6.2   | 5.5  |
|                                                                                  |             | MCWBR  | 4.9   | 4.7   | 4.2   | 4.7   | 5.1   | 6.8   | 6.4   | 5.9   | 5.5   | 5.7   | 5.6   | 5.2  |
|                                                                                  | 5% WB       | MCDBR  | 4.7   | 4.6   | 4.3   | 4.8   | 7.5   | 11.3  | 10.0  | 8.7   | 7.5   | 6.7   | 5.7   | 5.2  |
|                                                                                  |             | MCWBR  | 4.8   | 4.5   | 4.2   | 3.7   | 4.8   | 6.6   | 6.1   | 5.4   | 4.9   | 5.5   | 5.4   | 5.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.300  | 0.328 | 0.370 | 0.418 | 0.454 | 0.432 | 0.454 | 0.429 | 0.362 | 0.346 | 0.306 | 0.288 |      |
|                                                                                  | taud        | 2.047  | 2.047 | 1.971 | 1.911 | 1.966 | 2.183 | 2.158 | 2.227 | 2.401 | 2.271 | 2.148 | 2.032 |      |
|                                                                                  | Ebn at Noon | 552    | 698   | 762   | 784   | 782   | 812   | 782   | 776   | 790   | 686   | 551   | 459   |      |
|                                                                                  | Edh at Noon | 64     | 96    | 133   | 165   | 166   | 136   | 137   | 120   | 88    | 77    | 58    | 50    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.48   | 1.14  | 2.33  | 3.66  | 4.57  | 4.90  | 4.76  | 3.78  | 2.90  | 1.63  | 0.70  | 0.35  |      |
|                                                                                  | RadStd      | 0.08   | 0.28  | 0.39  | 0.49  | 0.48  | 0.54  | 0.41  | 0.33  | 0.22  | 0.13  | 0.06  | 0.05  |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page